



Embedded systems summary

Embedded Systems (Amrita Vishwa Vidyapeetham)



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Embedded System

Definition of an Embedded System

An embedded system is a computer system ,a combination of a computer processor, computer memory, and input/output peripheral devices—that has a dedicated function within a larger mechanical or electrical system.

Embedded means something that is attached to another thing.

Main components of an Embedded system

- Hardware
- Application software
- Real Time Operating System: RTOS defines the way the system works. It sets the rules during the execution of application programs.
- Small scale embedded system may not have RTOS.

Characteristics of an Embedded system

Single-functioned: An embedded system usually performs a specialized operation and does the same repeatedly.

Tightly-Constrained: All computing systems have constraints on design metrics but those on an embedded system can be especially tight.

Reactive and Real Time: Many embedded systems must continually react to changes in the system's environment and compute certain results in real time without any delay.

Memory: It must have a memory, as its usually embeds in ROM. It does not need any secondary memories in the computer.

Connected: it must have peripherals to connect input and output devices.

HW-SW systems: Software is used for features and flexibility. Hardware is used for performance and security.

Advantages & Disadvantages of an embedded system

- Advantages:
- Easily customizable.
- Low power consumption
- Low cost
- Enhanced performance.
- Disadvantages:
- high development effort.
- Lager time to market.

Structure of an Embedded System

Embedded system-Processor

What is a processor?

Main units of a Processor?

Types of Processor

GPP(General purpose Processor)

ASSP (Application Specific System Processor)

ASIP (Application Specific Instruction Processor)

A compiler is a computer program or a set of programs that transforms the source code written in a programming language the source language into another computer language normally binary format. The most common reason

for conversion is to create an executable program. The name "compiler" is primarily used for programs that translate the source code from a high level programming language to a low-level language e.g., assembly language or **machine code**.

What is Decompiler?

A program that can translate a program from a low-level language to a high-level language is called a decompiler

What is language converter?

A program that translates programs written in different high-level languages is normally called a language translator, source to source translator, or language converter.

A compiler is likely to perform the following operations –

- Preprocessing
- Parsing
- Semantic Analysis *Syntax-directed translation*
- Code generation
- Code optimization

What is an Assembler?

An assembler is a program that takes basic computer instructions called as assembly language and converts them into a pattern of bits that the computer's processor can use to perform its basic operations. An assembler creates object code by translating assembly instruction mnemonics into opcodes, resolving symbolic names to memory locations. Assembly language uses a mnemonic to represent each low-level machine operation opcode.

What is the importance of Debugging tools?

Debugging is a methodical process to find and reduce the number of bugs in a computer program or a piece of electronic hardware, so that it works as expected. Debugging is difficult when subsystems are tightly coupled, because a small change in one subsystem can create bugs in another. The debugging tools used in embedded systems differ greatly in terms of their development time and debugging features.

What are the Debugging tools you know in Embedded systems?

- Simulators
- Microcontroller Starter Kits
- Emulators

What is Simulator?

Code is tested for the MCU / system by simulating it on the host computer used for code development. Simulators try to model the behavior of the complete microcontroller in software.

What is the advantage of Microcontroller Starter Kits over Simulators?

Advantage of these kits over simulators is that they work in real-time and thus allow for easy input/output functionality verification. Starter kits, however, are completely sufficient and the cheapest option to develop simple microcontroller projects.

What is Emulation?

Emulation refers to the ability of a computer program in an electronic device to emulate imitate another program or device.

Emulation focuses on recreating an original computer environment.

Emulators have the ability to maintain a closer connection to the authenticity of the digital object.

An emulator helps the user to work on any kind of application or operating system on a platform in a similar way as the software runs as in its original environment.

What is the purpose of using Peripheral devices in Embedded systems?

Embedded systems communicate with the outside world via their peripherals

What is the criteria for choosing a Microcontroller?

While choosing a microcontroller, make sure it meets the task at hand and that it is cost effective.

We must see whether an 8-bit, 16-bit or 32-bit microcontroller can best handle the computing needs of a task.

Speed – What is the highest speed the microcontroller can support?

Packaging – Is it 40-pin DIP *Dual-inline-package* or QFP *Quad flat package*?

This is important in terms of space, assembling, and prototyping the end-product.

Power Consumption – This is an important criteria for battery-powered products.

Amount of RAM and ROM on the chip.

Count of I/O pins and Timers on the chip.

Cost per Unit – This is important in terms of final cost of the product in which the microcontroller is to be used.

Further, make sure you have tools such as compilers, debuggers, and assemblers, available with the microcontroller.

Little-Endian Vs Big-Endian

Although numbers are always displayed in the same way, they are not stored in the same way in memory.

Big-Endian machines store the most significant byte of data in the lowest memory address.

Little-Endian machines, on the other hand, store the least significant byte of data in the lowest memory address.

For e.g. for storing the data 0x12345678 is.....

Big-Endian machines

ADD+0: 0x12

ADD+1: 0x34

ADD+2: 0x56

ADD+3: 0x78

Little-Endian machines

ADD+0: 0x78

ADD+1: 0x56

ADD+2: 0x34

ADD+3: 0x12

What are the features of an 8051 Microcontroller?

64K bytes on-chip program memory *ROM*

128 bytes on-chip data memory *RAM*

Four register banks

128 user defined software flags

8-bit bidirectional data bus

16-bit unidirectional address bus

32 general purpose registers each of 8-bit

16 bit Timers *usually 2, but may have more or less*

Three internal and two external Interrupts

Four 8-bit ports, *short model have two 8-bit ports*

16-bit program counter and data pointer

8051 may also have a number of special features such as UARTs, ADC, Op-amp, etc.

PIC microcontroller is fast and easy to implement program when we compare other microcontrollers like 8051. The ease of programming and easy to interfacing with other peripherals PIC became successful microcontroller.

How can you justify PIC is fast compared to 8051?

PIC mainly used modified Harvard architecture and also supports RISC (Reduced Instruction Set Computer) by the above specification RISC and Harvard we can easily state that PIC is faster than the 8051 based controller which is made-up of Von-Newman architecture.

Advantages of PIC?

They are reliable and malfunctioning of PIC percentage is very less. And performance of the PIC is very fast because of using RISC architecture.

Power conception is also very less when compared to other micro controllers. When we see in the programmer point of view interfacing is very easy, also we can connect analog devices directly with out any extra circuitry and use them. Programming is also very easy when compared to other microcontrollers.

Disadvantages of PIC?

The length of the program will be big because of using RISC (35 instructions).

Program memory is not accessible and only one single accumulator is present.

Specific Interview Questions

1. What is watchdog timer?

A watchdog timer (or computer operating properly timer) is a computer hardware timing device that triggers a system reset if the main program, due to some fault condition, such as a hang, neglects to regularly service the watchdog. The intention is to bring the system back from the hung state into normal operation.

2. What is semaphore?

A semaphore is a protected variable or abstract data type which constitutes the classic method for restricting access to shared resources such as shared memory in a parallel programming environment. A counting semaphore is a counter for a set of available resources, rather than a locked/unlocked flag of a single resource.

3. What are hard and soft Real time systems?

The hard real time systems are the ones that depend on the output very strictly on time. Any late response or delay cannot be tolerated and will always be considered a failure. The soft real time systems on the other are not very rigid as the hard real time systems. The performance of the system degrades with the lateness of response, but it is bearable and can be optimized to a certain level for reuse of the result.

